

ThingSpeak Channel Reference Document

Mehrano Agri Farms (IoT Monitoring System)

PakAgri Digital Solutions

1. Channel Overview

This document provides a complete reference for the ThingSpeak channel used in the **Mehrano Agri Farms** IoT monitoring system. The channel collects real-time environmental and operational data from farm sensors simulated via **Cisco Packet Tracer** and transmitted through the system pipeline.

Property	Value
Channel Name	Mehrano Agri Farms Monitoring
Channel ID	3326236
Owner	PakAgri Digital Solutions
Platform	ThingSpeak by MathWorks
Channel URL	https://thingspeak.mathworks.com/channels/3326236
Total Fields	8
Data Range	April 1, 2026— April 13, 2026
Total Records	1727 entries
Data Interval	30 minutes per entry

2. API Keys

The following API keys are used to authenticate read and write operations on the ThingSpeak channel. The Write API Key is used by the Python data sender script and **Node-RED** flows. The Read API Key is used by **Power BI** to retrieve data for the management dashboard.

Key Type	Key Value	Usage
Write API Key	7WNCBDDVLZ8MKUO4	Sending sensor data to channel
Read API Key	T9HV37JVELHINJSS	Reading data (Power BI, dashboards)

3. Channel Fields

The channel contains 8 fields representing sensor readings and derived status values from the farm monitoring system. Fields 1 to 4 contain raw sensor readings. Fields 5 and 6 contain automated status values derived from sensor thresholds. Fields 7 and 8 contain human-readable condition labels.

Field	Name	Unit	Range	Description
Field 1	Soil Moisture	%	15 — 95	Soil water content percentage
Field 2	Temperature	°C	18 — 44	Air temperature reading
Field 3	Humidity	%	25 — 98	Relative humidity percentage
Field 4	Wind Speed	km/h	0 — 65	Wind speed measurement
Field 5	Pump Status	0 / 1	0 or 1	Irrigation pump state (1=ON, 0=OFF)
Field 6	Alert Status	0 / 1	0 or 1	System alert state (1=ACTIVE, 0=OK)
Field 7	Soil Condition	Label	Text	Critical / Low / Normal / Saturated
Field 8	Heat Stress	Label	Text	Heat Stress Risk / Optimal / Cool

4. Automated Threshold Logic

The following thresholds are applied automatically to derive status values in Fields 5 and 6. These rules simulate the automated decision-making logic of the IoT system.

Field	Condition	Result	Justification
Pump Status (F5)	Soil Moisture < 40%	1 (ON)	Soil too dry — irrigate
Pump Status (F5)	Soil Moisture >= 40%	0 (OFF)	Soil moisture sufficient
Alert Status (F6)	Temperature > 40°C	1 (ACTIVE)	Heat stress risk to crops

Alert Status (F6)	Temperature <= 40°C	0 (OK)	Temperature within safe range
Soil Condition (F7)	Soil Moisture < 20%	Critical	Immediate irrigation needed
Soil Condition (F7)	Soil Moisture 20-40%	Low	Irrigation recommended soon
Soil Condition (F7)	Soil Moisture 40-70%	Normal	Acceptable moisture level
Soil Condition (F7)	Soil Moisture > 70%	Saturated	No irrigation required

5. API Endpoints

The following REST API endpoints are used by the system components to interact with the ThingSpeak channel. These endpoints support both reading and writing sensor data.

5.1 Write Data (Single Entry):

Used by the Python simulation script to send individual sensor readings.

POST <https://api.thingspeak.com/update>

Parameters: `api_key`, `field1`, `field2`, `field3`, `field4`, `field5`, `field6`, `field7`, `field8`

Example:

https://api.thingspeak.com/update?api_key=7WNCBDDVLZ8MKUO4&field1=65.5&field2=32.1

5.2 Bulk Write (Multiple Entries):

Used by the bulk upload Python script to send up to 960 entries at once.

POST https://api.thingspeak.com/channels/3326236/bulk_update.json

Body: `{write_api_key: '7WNCBDDVLZ8MKUO4', updates: [{created_at, field1...field8}]}`

5.3 Read All Data (JSON):

Used by Power BI and external applications to retrieve stored sensor data.

GET

https://api.thingspeak.com/channels/3326236/feeds.json?api_key=T9HV37JVELHINJSS&results=1000

5.4 Read Latest Entry:

Retrieves the most recent sensor reading from the channel.

GET

https://api.thingspeak.com/channels/3326236/feeds/last.json?api_key=T9HV37JVELHINJSS

5.5 Read Single Field:

Retrieves data for one specific field only. Replace {fieldID} with 1 through 8.

GET https://api.thingspeak.com/channels/3326236/fields/{fieldID}.json?api_key=T9HV37JVELHINJSS&results=100

Example (Soil Moisture):

```
https://api.thingspeak.com/channels/3326236/fields/1.json?api_key=T9HV37JV  
ELHINJSS
```

5.6 Channel Status:

Returns current channel metadata including field names and last update time.

GET

```
https://api.thingspeak.com/channels/3326236/status.json?api_key=T9HV37JVEL  
HINJSS
```

6. JSON Response Format

The following shows the structure of a typical JSON response from the ThingSpeak Read API.

This format is used by Power BI when connecting to the channel via the Web connector.

```
{  
  "channel": {  
    "id": 3326236,  
    "name": "Mehrano Agri Farms Monitoring",  
    "field1": "Soil Moisture",  
    "field2": "Temperature",  
    "field3": "Humidity",  
    "field4": "Wind Speed",  
    "field5": "Pump Status",  
    "field6": "Alert Status",  
    "field7": "Soil Condition",  
    "field8": "Heat Stress"  
  },  
  "feeds": [  
    {  
      "created_at": "2026-03-01T00:00:00Z",  
      "entry_id": 1,  
      "field1": "65.32",  
      "field2": "28.74",  
      "field3": "78.12",  
      "field4": "12.45",  
      "field5": "0",  
      "field6": "0",  
      "field7": "Normal",  
      "field8": "Optimal"  
    }  
  ]  
}
```

7. Power BI Connection Details

Power BI connects to ThingSpeak using the Web connector with the Read API Key. The following settings are used to establish the connection.

Setting	Value
Connector Type	Web (URL)
Base URL	https://api.thingspeak.com/channels/3326236/feeds.json
API Key Parameter	api_key=T9HV37JVELHINJSS
Results Parameter	results=1000
Full URL	https://api.thingspeak.com/channels/3326236/feeds.json?api_key=T9HV37JVELHINJSS&results=1000
Data Format	JSON
Refresh Method	Manual / Scheduled

8. Node-RED Integration (Iteration 2)

From Iteration 2 onwards, Node-RED acts as the middleware layer between the MQTT broker and ThingSpeak. The following settings are used to configure the Node-RED ThingSpeak node.

Setting	Value
Node-RED Version	3.x
ThingSpeak Node	ThingSpeak Upload (HTTP)
Dashboard Node	node-red-dashboard
MQTT Broker IP	192.168.25.101
MQTT Port	1883
MQTT Topics	farm/soil, farm/temperature, farm/humidity, farm/pump
ThingSpeak Write Key	7WNCBDDVLZ8MKUO4
Node-RED URL	http://127.0.0.1:1880